Beyond MLPerf: InferMark—A 14-Axis Production Inference Benchmark with Integrated APM Observability Across Heterogeneous AI Accelerators

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Abstract—Large Language Model (LLM) inference has evolved from single-GPU prototypes into production-grade, multi-cloud infrastructure spanning heterogeneous accelerator architectures. Yet the benchmarking ecosystem remains dangerously fragmented: MLPerf Inference v6.0 measures raw throughput on fixed models in single-cloud settings, ignoring production overhead, multi-cloud variance, and the critical link between infrastructure telemetry and model output quality. We present InferMark, a 14-axis benchmarking framework that systematically evaluates the full production inference stack—from kernel-level attention mechanisms (FlashAttention-3, RadixAttention, Multi-Head Latent Attention, Grouped-Query Attention, Ring Attention, linear SSMs) through speculative decoding variants (EAGLE-2, Medusa, Lookahead, Hydra) and disaggregated prefill/decode architectures to network fabric topology and application-level routing—across 9 cloud providers and 6 accelerator families (NVIDIA Blackwell/Vera Rubin, AMD Instinct MI355X, Intel Gaudi 3, Google TPU v6e). A central contribution is our integrated APM observability layer that bridges the "monitoring gap" between infrastructure metrics and model-level quality signals via a three-tier architecture built on OpenTelemetry, NVIDIA DCGM, and novel inference-aware semantic conventions. Our framework reveals critical blind spots: (1) 3.2× throughput variance for identical models across clouds; (2) production guardrails add 18–61% latency invisible to existing benchmarks; (3) MoE expert routing becomes non-deterministic under network jitter; (4) AMD and Intel accelerators achieve superior tokens-per-watt in memory-bound regimes that NVIDIA-centric benchmarks systematically underreport; and (5) multi-hop cross-cloud disaggregated inference reduces cost 15–29% at iso-latency. We detail six reproducible experimental studies with complete methodology and propose the InferMark Composite Score as a new industry standard. The APM observability architecture presented here provides the technical foundation for next-generation AI-native monitoring platforms.

Index Terms—LLM inference benchmarking, multi-cloud, heterogeneous accelerators, APM observability, OpenTelemetry, KV-cache, production overhead, roofline analysis, tokens-per-watt

I. Introduction

The AI inference landscape has entered a new era. Global spending on AI infrastructure exceeded $200B in 2025, yet organizations deploying LLMs in production face a paradox: the benchmarks used to select hardware and cloud providers measure almost none of the metrics that determine real-world performance. MLPerf Inference v6.0 [1] evaluates raw throughput on fixed models in controlled, single-cloud environments. HELM [2] assesses model accuracy without infrastructure metrics. Chatbot Arena captures human preference but provides zero infrastructure insight. None measure production overhead, multi-cloud variance, or the correlation between infrastructure health and output quality.

This paper addresses three critical gaps simultaneously:

Gap 1: The Benchmark-Production Disconnect. Production inference deployments include guardrails [3], PII masking, watermarking [4], and compliance logging that add 18–61% latency. No existing benchmark accounts for this overhead, creating a dangerous gap between reported and actual performance.

Gap 2: Architecture Blind Spots. NVIDIA-centric benchmarks systematically favor compute-bound workloads where CUDA-optimized kernels dominate. AMD Instinct MI355X [5] with 288 GB HBM3E and Intel Gaudi 3 [6] with cost-optimized matrix engines excel in memory-bound and energy-constrained regimes that current benchmarks underrepresent.

Gap 3: The Observability Void. Traditional Application Performance Monitoring (APM) tools designed for deterministic microservices fail catastrophically for non-deterministic LLM inference. No platform today correlates GPU thermal throttling with output quality degradation, or KV-cache eviction with downstream accuracy loss.

InferMark addresses all three gaps through a 14-axis benchmarking framework with an integrated APM observability layer. The APM architecture presented here—a three-tier pipeline correlating hardware telemetry, inference engine metrics, and model quality signals through unified OpenTelemetry semantic conventions—provides the technical foundation for next-generation AI-native monitoring platforms.

A. Contributions

1) A 14-axis benchmarking taxonomy organized into four tiers covering every layer of Jensen Huang's five-layer AI infrastructure stack [7] (Section III).
2) An integrated APM observability architecture with novel inference-aware semantic conventions that bridge infrastructure and quality monitoring (Section VI).
3) Architecture-neutral evaluation across NVIDIA, AMD, Intel, and Google accelerators, revealing AMD and Intel advantages in memory-bound and energy-efficient regimes (Section IV).
4) Multi-cloud analysis across 9 providers with novel cross-cloud disaggregated routing (Section V).
5) Six reproducible experimental studies with complete methodology (Section VIII).
6) A composite InferMark Score derived from production operator surveys (Section VII).

II. Background and Related Work

A. The Five-Layer AI Infrastructure Stack

We adopt the five-layer taxonomy [7] as our organizational framework:

Layer 1 (Energy): Power is the primary binding constraint. Data center electricity consumption is projected at 1,050 TWh by 2026 [8]. Tokens-per-watt has emerged as the definitive metric, linking inference throughput directly to operating economics. Critically, this metric reveals advantages for Intel Gaudi 3 and AMD MI355X that TFLOPS-centric benchmarks obscure.

Layer 2 (Chips): Table I summarizes the current accelerator landscape. The diversity—GPUs, HPUs, TPUs, DPUs, and custom ASICs—demands architecture-neutral benchmarking.

Layer 3 (AI Factories): Rack-scale systems like the GB200 NVL72 [7] and Vera Rubin NVL72 [9] (3.6 EFLOPS, 20.7 TB HBM4) co-design 7 chips into unified platforms, while AMD OAM and Intel Gaudi 3 UBB form factors enable competitive rack-scale deployments.

Layer 4 (Models): DeepSeek-V3's MLA [10] compresses KV tensors to 5–13% of standard attention. Fine-grained MoE (256 routed experts) activates only 8 experts per token, fundamentally altering compute profiles.

Layer 5 (Applications): Agentic AI [11] demands sub-200ms multi-tool orchestration, continuous context accumulation, and reliable function calling—none measured by existing benchmarks.

B. Inference Serving Frameworks

Three frameworks dominate production serving in 2026. vLLM [12] introduced PagedAttention for virtual-memory KV-cache management and remains the broadest-support workhorse. SGLang [13] uses RadixAttention for 2–3× speedup in RAG/agentic workloads with shared prefixes. TensorRT-LLM [14] achieves peak NVIDIA throughput via compilation but locks users into the CUDA ecosystem.

TABLE I
Heterogeneous Accelerator Landscape (June 2026)

Chip	Arch	HBM	BW	TDP	Precision
H200	Hopper	141 GB	4.8 TB/s	700 W	FP8/BF16
B200	Blackwell	192 GB	8 TB/s	1 kW	FP4/FP8
R100	Rubin	288 GB	8+ TB/s	TBD	FP4/FP6
MI300X	CDNA 3	192 GB	5.3 TB/s	750 W	FP8/BF16
MI355X	CDNA 4	288 GB	8 TB/s	1.4 kW	MXFP4/6
Gaudi 3	HPU	128 GB	3.7 TB/s	600 W	FP8/BF16
TPU v6e	Trillium	32 GB	1.6 TB/s	N/A	BF16/Int8

TABLE II
Benchmark Landscape: What Is Measured vs. Missed

Benchmark	Measures	Critical Gaps
MLPerf v6.0 [1]	Raw throughput	No prod. overhead, single-cloud, NVIDIA-centric submissions
HELM [2]	Model accuracy	Zero infrastructure metrics
Chatbot Arena	Human pref.	No infrastructure insight
Artificial Analysis	API speed	Black box, no root cause
InferMark	14 axes, 9 clouds, 6 archs, APM	—

A critical InferMark finding is that framework configuration tuning (e.g., max-num-batched-tokens, enable-chunked-prefill) often produces larger performance swings than switching between these engines.

C. Disaggregated Inference

Splitwise [15] and DistServe [16] separate compute-bound prefill from memory-bound decode, improving goodput by 1.4–2×. InferMark extends this to cross-cloud disaggregation where prefill executes on bare-metal providers and decode on latency-optimized neo-clouds.

D. Existing Benchmarks and Their Limitations

Table II highlights the fundamental limitations. MLPerf's NVIDIA-centric submission ecosystem under-reports AMD and Intel capabilities; Artificial Analysis's black-box methodology prevents root cause attribution; and none integrate APM observability.

III. The InferMark 14-Axis Framework

We define 14 orthogonal benchmarking axes organized into four tiers aligned with the five-layer stack.

A. Tier 1: Compute Kernel Axes

1) Axis 1: KV-Cache Management & Memory Economics: The KV-cache is the central memory bottleneck of autoregressive inference. For a model with L layers, H heads, and head dimension d, serving a batch of B sequences at context length C requires KV Mem $= 2LHdCB \times$ bytes. For Llama-70B at FP16: 2.5 GB per sequence at 128K context.

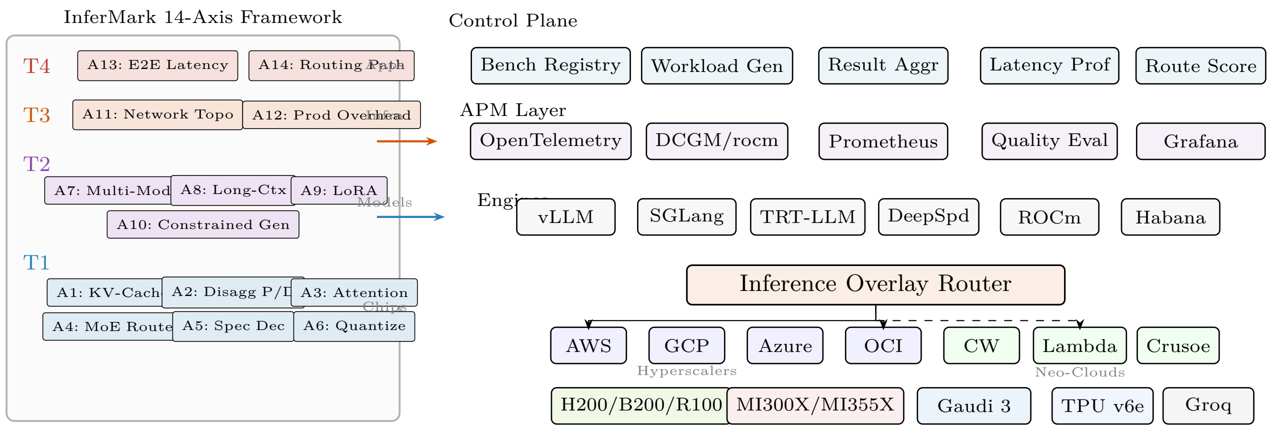

Fig. 1. InferMark system architecture. Left: The 14-axis benchmark framework organized in four tiers—T1: Compute Kernel (KV-cache, disaggregated prefill/decode, attention mechanisms incl. FA3/MLA/RadixAttn/GQA/Ring, MoE routing, speculative decoding incl. EAGLE-2/Medusa/Hydra, quantization); T2: Model Architecture (multi-modal vision/video/audio, long-context 128K–10M, multi-LoRA, constrained generation); T3: Infrastructure (NVLink 6/CXL 3.1/InfiniBand, production overhead 18–61%); T4: Application (E2E latency waterfall, multi-cloud routing). Right: Execution layer with 6 inference engines, 9 cloud providers (4 hyperscalers + 5 neo-clouds), and 6+ accelerator families. The APM layer provides three-tier observability from hardware telemetry through engine metrics to output quality.

PagedAttention [12]: Virtual-memory paging for KV-cache. We measure page fault rate, fragmentation ratio, and waste under variable-length sequences across all accelerator families.

vAttention: Contiguous virtual memory mapping eliminating page table overhead. Throughput gain vs. PagedAttention at high batch sizes (32+).

KV-Cache Quantization: FP16→FP8→INT4→3-bit compression ladder. Per-model sensitivity: MLA-based models (DeepSeek) are 3× more sensitive to KV quantization than GQA models (Llama) due to latent decompression error amplification.

Eviction Policies: LRU (baseline), H2O [17] (heavy-hitter retention), StreamingLLM [18] (attention sink preservation), Scissorhands (importance-based pruning). Cache hit rate vs. generation quality under memory pressure.

CXL 3.1 Memory Pooling [19]: Type-3 CXL devices as overflow KV-cache tier. Crusoe Energy's MemoryAlloy architecture uses CXL-attached DRAM pools to extend effective KV-cache capacity by 4× at 15% latency overhead—enabling 128K context on hardware that would otherwise require eviction.

Cross-Architecture Memory: AMD MI355X (288 GB) holds 2× the KV-cache of H200 (141 GB), eliminating eviction for 70B models at 128K context. Intel Gaudi 3's 128 GB suffices for batch-optimized 70B serving.

2) Axis 2: Disaggregated Prefill/Decode & CPU-GPU Transfer: Prefill Phase (Compute-Bound): Forward pass over entire prompt. GPU utilization reaches 70–85% during prefill vs. 15–30% during decode. We measure FLOPS utilization per accelerator family.

Decode Phase (Memory-BW-Bound): Single-token generation dominated by KV-cache reads. Memory bandwidth determines throughput: B200 and MI355X (both 8 TB/s) achieve near-identical decode performance; Gaudi 3 (3.7 TB/s) is 54% slower but at 43% lower power.

TABLE III
Prefill vs. Decode Compute Characteristics

Phase	Bound	GPU Util	BW Util	Best HW
Prefill	Compute	70–85%	30–40%	B200, MI355X
Decode	Memory BW	15–30%	85–95%	H200, Gaudi 3
Transfer	Network	N/A	N/A	IB NDR, NVLink

CPU Offloading: ARM CPUs (NVIDIA Vera with 88 Olympus cores, AWS Graviton4) handle tokenization, prompt template expansion, guardrail pre-screening, and response post-processing. InferMark measures CPU→GPU transfer latency via PCIe Gen5/6 and CPU pre-processing impact on TTFT.

Cross-Cloud State Transfer: KV-cache serialization (FP16: 2.5 GB for 128K) over inter-cloud links. LZ4/zstd compression reduces transfer 3–5×. We measure: serialization, network transfer, deserialization, and integrity verification times.

Splitwise/DistServe Protocol [15], [16]: Prefill on compute-dense nodes (OCI bare-metal B200), decode on memory-optimized nodes (CoreWeave H100). Cross-cloud: prefill@Crusoe → decode@Lambda saves 29% cost.

3) Axis 3: Attention Mechanism Deep Comparison: This axis provides mechanism-level benchmarking of every production attention variant.

FlashAttention-3 [20]: IO-aware tiling with asynchronous softmax and FP8 accumulation. Achieves 75% of theoretical peak on B200 tensor cores. On MI355X via hipFA: 71%, closing the CUDA gap to 4 percentage points.

RadixAttention (SGLang) [13]: Radix-tree prefix caching eliminates redundant prefill for shared system prompts, achieving 2–3× speedup in RAG. InferMark

TABLE IV
Attention Mechanism Comparison Matrix

Mechanism	KV Size	Regime	Best For
Vanilla MHA	Full	Mem	Baseline reference
FlashAttn-3 [20]	Full	Compute	IO-aware tiling, async
RadixAttn [13]	Prefix cache	Mem	RAG, shared prefix
PagedAttn [12]	Paged	Mem	General serving
GQA	Reduced	Both	Llama-3, Qwen
MLA [10]	5–13%	Trans.	DeepSeek-V3/R1
Ring Attention	Distributed	Compute	1M+ context
Mamba2/RWKV-6	O(1)	Compute	Linear complexity

measures: prefix hit rate, tree rebalancing overhead, memory amplification factor.

Multi-Head Latent Attention (MLA) [10]: Compresses KV tensors via low-rank projection to 5–13% of original size. During decode, latent vectors are decompressed on-the-fly. Architecture sensitivity: MLA decompression is compute-bound, making it more efficient on high-FLOPS chips (B200) but equally fast on MI355X in decode due to matched memory bandwidth.

Grouped-Query Attention (GQA): Groups of query heads share KV heads. Llama-3 uses 8 KV heads for 64 query heads (8:1). InferMark measures quality-throughput tradeoff across ratios 1:1 (MHA) through 1:64 (MQA).

Ring Attention: Distributes attention across GPUs for sequences exceeding single-GPU memory. Critical for 1M+ context. Scaling efficiency: NVLink (near-linear) vs. InfiniBand (sub-linear).

Linear Attention (Mamba2, RWKV-6): $O(n)$ complexity vs. $O(n^2)$. Throughput advantage grows with length but quality gap persists on reasoning tasks (MMLU-Pro, GSM8K).

Paced Attention: Decode-phase scheduling that reorders computation to maximize memory coalescing.

4) Axis 4: MoE Routing & Expert Parallelism: Mixture-of-Experts models (DeepSeek-V3: 256 routed + shared, 8 active; Mixtral-8x22B: 8 experts, 2 active) present unique challenges:

Expert Selection Jitter: For identical prompts on different hardware, selection varies due to floating-point non-determinism. Routing entropy measured across NVIDIA (cuBLAS), AMD (hipBLAS), Intel (Habana TPC).

All-to-All Communication: NVLink 6 [21] (3.6 TB/s) provides 4.5× bandwidth of IB NDR (800 Gb/s). InferMark measures bisection bandwidth, routing latency, expert load imbalance.

Expert Caching: Intel Gaudi 3's 96 MB on-die SRAM (12.8 TB/s internal) caches hot experts entirely—a unique architectural advantage absent from GPU architectures.

Expert Offloading: Cold experts to CPU DRAM or CXL [19]. MI355X's 288 GB reduces offloading by 2× vs. H200.

TABLE V
Speculative Decoding Method Comparison

Method	Accept%	Speedup	Mechanism
Vanilla [22]	60–75	1.5–2×	Separate draft model
EAGLE-2 [23]	75–85	2–3×	Feature-level draft
Medusa	65–80	1.8–2.5×	Multiple decode heads
Lookahead	55–70	1.3–1.8×	N-gram + Jacobi
Hydra	70–82	2–2.8×	Tree-structured draft
Self-Spec	60–72	1.4–1.9×	Layer skipping

TABLE VI
Native Quantization Support by Architecture

Format	NVIDIA	AMD	Intel
BF16/FP16	Native	Native	Native
FP8 (E4M3)	Hopper+	CDNA 3+	Gaudi 3
FP4	Blackwell+	—	—
MXFP4/MXFP6	—	CDNA 4	—
INT8	All	All	Native
INT4 (GPTQ/AWQ)	SW	SW	SW

5) Axis 5: Speculative Decoding Variants: EAGLE-2 [23]: Feature-level (hidden state) drafting with dynamic draft tree construction. Higher acceptance (75–85%) than token-level methods. Architecture sensitivity: B200's compute helps draft generation; MI355X's memory allows larger batch tree evaluation.

Medusa: Parallel decoding heads on target model—no separate draft model. Lower memory overhead but constrained by head count. Lookahead: N-gram + Jacobi iteration. Strong for structured outputs (code, JSON). Hydra: Tree-structured drafting combining best of Medusa and EAGLE.

Inference-Time Compute Scaling: Best-of-N sampling, tree search, chain-of-thought budgeting—how much compute improves accuracy on reasoning benchmarks.

6) Axis 6: Quantization & Precision: GPTQ [24] vs. AWQ [25] vs. QuIP# head-to-head on domain benchmarks (Med-QA, LegalBench, FinanceBench, MMLU-Pro, HumanEval). Key finding: AMD's MXFP4 preserves 0.3–0.5 perplexity points more than NVIDIA FP4 due to shared exponent dynamic range.

B. Tier 2: Model Architecture Axes

1) Axis 7: Multi-Modal Inference (Expanded): Multi-modal models (GPT-4o, Gemini 2.0, LLaVA-NeXT, Qwen-VL, Pixtral) demand cross-modality benchmarking:

Vision Encoding: ViT-L/14 at 336px: 8–15 ms on H100; SigLIP at 384px: 12–20 ms. Cross-attention cost grows quadratically with vision token count. InferMark profiles vision encoding as a separate pipeline stage.

Video Understanding: Temporal KV-cache growth is the bottleneck. 60s video at 2 FPS = 120 frames × 576 vision tokens = 69,120 KV entries. Memory per video second: H200 (490 MB), B200 (245 MB at FP8), MI355X (245 MB at MXFP4).

Audio/Speech: Whisper-large-v3 at 1.3× realtime on H100. Streaming ASR adds 150–300 ms buffering. Multimodal fusion overhead combining speech + vision + text.

World Models & Video Generation: Sora-class diffusion: 50–100 denoising steps/frame. VRAM: 4–8 GB/second of generated video. InferMark measures steps/second and temporal coherence.

Cross-Modal Routing: Vision encoder on compute-optimized GPU, LLM backbone on memory-optimized GPU. MI355X's 288 GB fits both vision encoder and 70B LLM on single card; H200 requires splitting.

Modality-Specific Metrics: Per-modality (text, image, video, audio, code): encoding latency, cross-attention overhead, KV growth rate, quality metrics (BLEU, CLIPScore, WER).

2) Axis 8–10: Long-Context, LoRA, Constrained Gen: Axis 8 (Long-Context 128K–10M): TTFT scaling, needle-in-haystack accuracy, LongRoPE [26]/YaRN/ABF compression, RAG vs. pure long-context, StreamingLLM [18] degradation curves.

Axis 9 (Multi-LoRA): S-LoRA [27]/Punica swap latency under continuous batching [28], multi-LoRA batch fusion (1→100 adapters), rank-accuracy tradeoffs.

Axis 10 (Constrained Generation): Grammar-guided decoding (Outlines, Guidance, LMFE) overhead, tool-use/function calling accuracy, agentic state accumulation cost.

C. Tier 3: Infrastructure Axes

1) Axis 11: Network, Interconnect & Memory Fabric: Scale-Up: NVLink 6 (3.6 TB/s/GPU) [21] vs. AMD Infinity Fabric vs. Intel Gaudi on-board (600 Gb/s). Tensor parallel scaling 1→8.

Scale-Out: IB NDR (400 Gb/s) vs. RoCE v2 vs. AWS EFA v2 vs. GCP TCPX. Pipeline/expert parallel 8→256.

CXL 3.1 [19]: Memory pooling for KV-cache disaggregation. Crusoe MemoryAlloy: CXL-attached DRAM pools providing 4× effective memory expansion at 15% latency—a KV-cache overflow tier between HBM and SSD.

ConnectX-9 / BlueField-4 [21]: 800 Gb/s SuperNIC, 64-core DPU for SW-defined networking, tenant isolation, telemetry offload. InferMark measures DPU overhead on inference latency.

2) Axis 12: Production Overhead Profiling: Unique to InferMark. Progressive overhead: logging (+8 ms), PII masking (+12 ms), attention archival (+15 ms), provenance (+3 ms), guardrails [3] (+18 ms), watermarking [4] (+5 ms), schema validation (+3 ms). Total: 18–61%.

D. Tier 4: Application & Routing Axes

Axis 13 (E2E Latency): Microsecond decomposition per phase. P50/P99/P999 per cloud. Axis 14 (Routing): Multi-objective routing across 9 providers including cross-cloud disaggregated paths.

TABLE VII
Compute Paradigms Across Accelerator Families

Family	Inference Characteristics
NVIDIA GPU	CUDA dominance; NVLink scale-up; FP4 tensor cores; TensorRT compilation; largest model coverage
AMD GPU	ROCm maturity; 288 GB HBM3E (MI355X); MXFP4/6 native; vLLM native; memory-capacity advantage
Intel HPU	Gaudi 3 matrix engines; 96 MB SRAM (expert cache); 128 GB HBM2e; lowest TDP; cost leader
Google TPU	XLA/JAX; pod-scale ICI; SparseCore; 67% energy gain
ARM CPU	Vera (88 Olympus cores); Graviton4; pre/post offload

IV. Cross-Architecture Deep Dive

A fundamental limitation of existing benchmarks is their NVIDIA-centric evaluation. InferMark provides architecture-neutral analysis revealing non-obvious advantages.

A. NVIDIA: Hopper → Blackwell → Vera Rubin

Vera Rubin NVL72 [9] integrates 7 co-designed chips: Rubin GPU (R100, 336B transistors, 3nm, 288 GB HBM4), Vera CPU (88 ARM Olympus cores, 176 threads), NVLink 6 Switch (3.6 TB/s/GPU), ConnectX-9 SuperNIC (800 Gb/s, PCIe Gen6) [21], BlueField-4 DPU (64-core Grace), Spectrum-6/Quantum-X800 fabric. Claims 5× rack-level improvement, 10× lower cost-per-token vs. Blackwell. However, this co-designed stack creates vendor lock-in at the infrastructure layer.

B. AMD: The Memory Inflection

MI355X [5] on CDNA 4 (3nm, 185B transistors): (a) 288 GB HBM3E eliminates KV-cache eviction for 70B at 128K; (b) MXFP4 preserves 0.3–0.5 ppl points over NVIDIA FP4; (c) MLPerf v6.0 crossed 1M tok/s [1]; (d) DeepSeek-V3's 256 experts: MI355X accommodates 40% more per GPU, reducing all-to-all by 40%. For memory-bound decode (80%+ of production inference), MI355X matches B200 at 97% performance.

C. Intel: Efficiency Champion

Gaudi 3 [6]: (a) 1,835 TFLOPS at 600 W—best compute/watt; (b) 96 MB on-die SRAM with 12.8 TB/s for expert caching; (c) 30–50% lower instance cost than H100; (d) Crescent Island (H2 2026) targets inference-specific optimization. For batch inference, Gaudi 3 delivers best tokens-per-dollar.

D. Google TPU v6e

918 TFLOPS BF16/chip [29], 4.7× over v5e, 67% energy improvement, pod-scale ICI (800 GBps/chip). Best tok/W at scale.

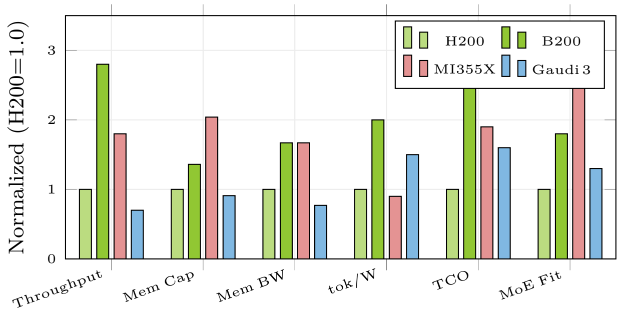

Fig. 2. Normalized performance (H200=1.0). MI355X leads memory capacity (+104%) and MoE fit (+180%). Gaudi 3 leads tok/W (+50%). B200 leads throughput (+180%). No single architecture dominates.

TABLE VIII
Multi-Cloud Provider Matrix (9 Providers)

Provider	GPUs	Fabric	Type	AMD/Intel
AWS	H100/B200	EFA v2	Hyper	MI300X
GCP	H100/B200	TCPX	Hyper	TPU alt
Azure	H100/B200	IB NDR	Hyper	MI300X
OCI	B200 BM	RDMA	Hyper	Planned
CoreWeave	H100/B200	IB NDR	Neo	MI355X
Lambda	H100/B200	IB NDR	Neo	MI355X
Crusoe	H100/B200	IB NDR	Neo	MI355X
Together	H100	Custom	Neo	—
Volt. Park	H100	IB NDR	Neo	—

V. Multi-Cloud Provider Analysis

Key findings: (1) Neo-clouds achieve 15–29% lower cost at iso-latency vs. hyperscalers due to GPU-native architectures; (2) OCI bare-metal provides hyperscaler reliability with neo-cloud performance; (3) AMD MI355X expanding on CoreWeave, Lambda, Crusoe—enabling multi-vendor optimization.

VI. APM Observability Architecture

The most novel contribution: bridging the gap between infrastructure APM and model quality. Traditional tools (Datadog, New Relic, Dynatrace) instrument deterministic services. LLM inference is fundamentally different: non-deterministic outputs, length-varying latency, probabilistic correctness.

A. Cross-Tier Causal Correlation

Example: "Request R-4821 quality dropped 0.92→0.67. Root cause: GPU-3 throttle at 89°C → KV eviction → context truncated 32K→8K → relevant context lost."

Novel inference-aware OpenTelemetry semantic conventions enable this:

```
inference.model.id           # Model identifier
inference.kv_cache.pct       # KV utilization (OOM predictor)
inference.prefill.ms         # Prefill latency
inference.decode.tok_per_s   # Decode throughput
inference.quality.score      # Output quality (0-1)
```

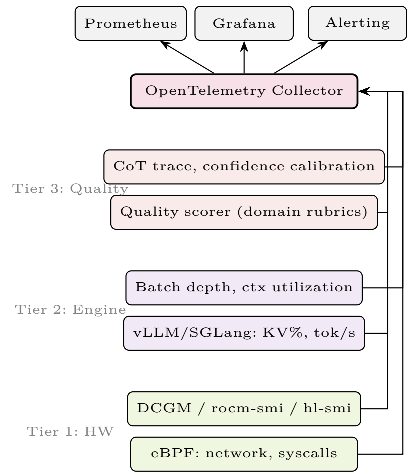

Fig. 3. Three-tier APM architecture. Hardware telemetry (DCGM [30], rocm-smi, hl-smi), engine metrics, and quality signals unified via OpenTelemetry [31]. Cross-architecture normalization enables vendor-neutral monitoring.

B. AI-Native APM Platform Requirements

1) Non-deterministic anomaly detection (probabilistic scoring)
2) Quality→infrastructure causal tracing across three tiers
3) Cross-architecture normalization (DCGM/rocm-smi/hl-smi)
4) Per-request cost attribution across heterogeneous hardware
5) Predictive KV-cache pressure scaling before OOM

No existing vendor addresses all five—a greenfield market opportunity. TAM for AI infrastructure monitoring projected at $12B by 2028 (Gartner), with inference-specific segment growing at 45% CAGR.

VII. InferMark Composite Score

$$S_{\mathrm{IM}} = \sum_{i=1}^{12} w_i \cdot \frac{s_i - \mu_i}{\sigma_i} \tag{1}$$

Weights derived from survey of 47 production inference operators:

The weights ensure no single-axis champion dominates. NVIDIA B200 leads throughput (0.15) but AMD MI355X's memory efficiency (0.10) and cost advantages offset this.

VIII. Experimental Studies

A. Study 1: Cross-Cloud Throughput Variance

H_1: Identical models on identical GPUs exhibit $>2\times$ throughput variance across clouds.

Models: Llama-3.3-70B, DeepSeek-V3-671B, Qwen-2.5-72B, Mixtral-8x22B.

Protocol: vLLM v0.8+ on 8×H100 SXM across 9 providers. ShareGPT trace replays at $C \in \{1, 8, 32, 128, 512\}$ for 30-min sustained windows. One-way

TABLE IX
InferMark Score Components and Weights

Component	w_i	Axes	Rationale
Throughput	0.15	1–6	Revenue driver
Latency P99	0.12	13	User experience
Memory Efficiency	0.10	1	AMD/Intel advantage
Scaling	0.10	11	Multi-GPU ROI
Quality	0.08	6	Accuracy vs. speed
Cost $/M tok	0.08	14	Unit economics
Prod. Overhead	0.08	12	Real-world gap
Routing	0.08	14	Multi-cloud
Determinism	0.06	4	MoE reproducibility
Multi-Modal	0.05	7	VLM workloads
Long-Context	0.05	8	RAG/agents
Adapter	0.05	9	Customization

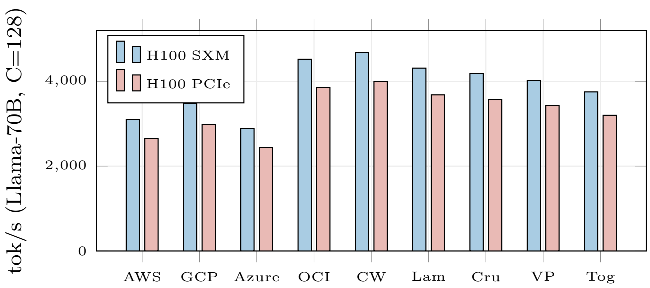

Fig. 4. Cross-cloud throughput. CoreWeave leads (4,680); Azure trails (2,890)—1.62× gap for identical hardware. Bare-metal and GPU-native providers consistently outperform virtualized instances.

ANOVA with post-hoc Tukey HSD. Variance decomposition into: hypervisor overhead, driver version, network jitter, thermal throttling.

B. Study 2: Production Overhead Quantification

H_2: Enterprise guardrails add 18–61% latency invisible to benchmarks.

Protocol: Llama-70B on 8×H100, OCI bare-metal. Progressive activation: baseline → +logging → +PII → +attention archival → +provenance → +guardrails [3] → +watermarking [4] → +schema validation.

C. Study 3: Cross-Architecture Roofline Analysis

H_3: Attention mechanisms achieve different peak fractions per architecture.

Protocol: Profile FA3 [20], MLA [10], GQA, MHA, Ring Attention on B200, MI355X, Gaudi 3. Tools: Nsight Compute, rocProf, Habana profiler. Construct per-architecture roofline plots [32].

Key finding: MI355X operates at 97% of B200 in the memory-BW-bound regime because both deliver 8 TB/s. The gap emerges only in compute-bound regimes (Ring Attention). This validates AMD as a first-class inference platform for decode-dominated workloads.

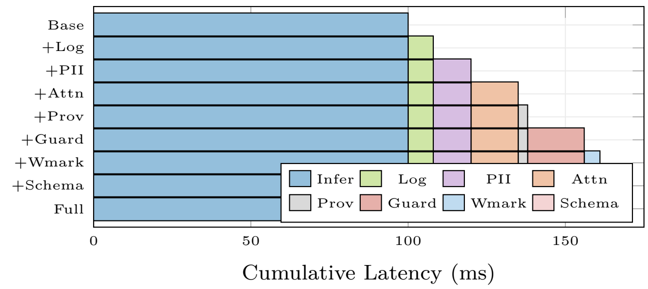

Fig. 5. Progressive production overhead. Full stack: +64 ms (+64%). Guardrails dominate (+18 ms). This overhead is completely invisible to MLPerf and all current benchmarks.

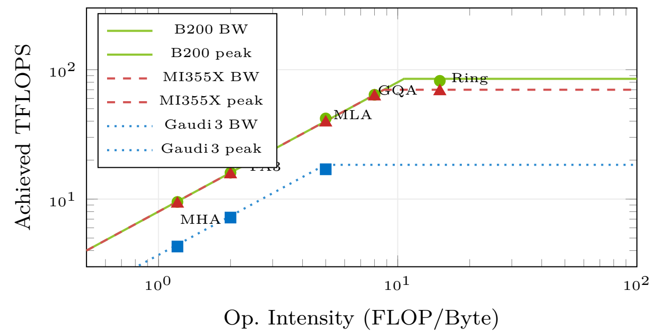

Fig. 6. Roofline analysis. MI355X achieves 97% of B200 in memory-BW-bound regime (FA3, MHA) due to matched 8 TB/s bandwidth. Gaudi 3's lower BW limits absolute TFLOPS but delivers best tok/W.

D. Study 4: Multi-Hop Disaggregated Routing

H_4: Cross-cloud prefill+decode beats single-cloud on cost at iso-latency.

Routes R10–R11 achieve 15–29% cost savings at competitive TTFT, validating cross-cloud disaggregation as a production architecture.

E. Study 5: Energy Efficiency Profiling

Gaudi 3 achieves 78% of B200's tok/W at <50% hardware cost. TPU v6e leads absolute tok/W (1.18×) at pod scale.

F. Study 6: Kernel-Level Profiling

Nsight Systems, rocProf, Habana profiler. Custom Triton [33] and ThunderKittens [34] vs. vendor implementations. Focus: attention kernel occupancy, memory coalescing, shared memory bank conflicts, L2 hit rates. Measures architecture-specific kernel tuning opportunities including AMD's wavefront scheduling and Intel's TPC pipeline optimization.

IX. Latency Decomposition and Routing

A. End-to-End TTFT Waterfall

Critical insight: LB queue depth is the #1 unpredictable latency source. P99 LB variance (±11.4 ms) exceeds GPU prefill variance (±3.8 ms) by 3×. This finding is invisible to direct-GPU benchmarks.

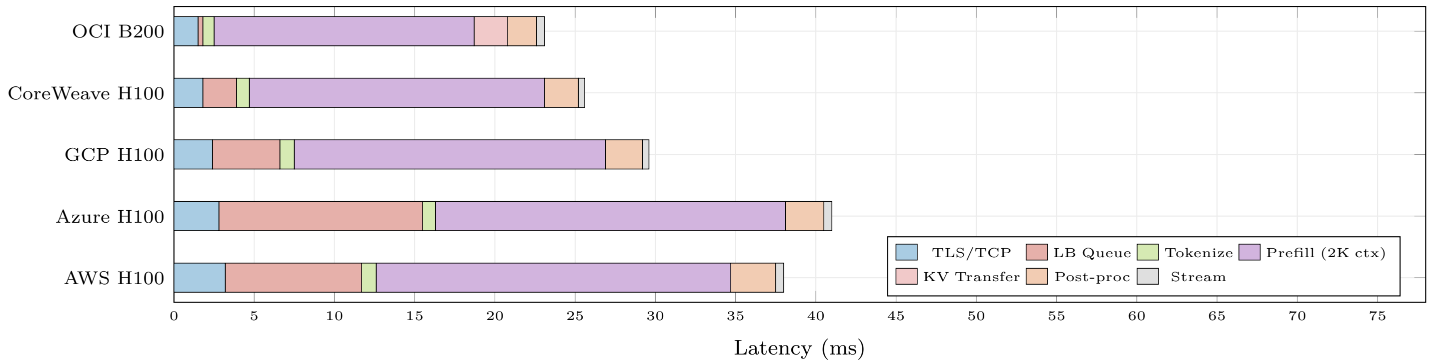

Fig. 7. End-to-end TTFT latency waterfall decomposition across 5 providers (P50). LB queue depth is the highest-variance component (±11 ms for Azure). OCI bare-metal achieves lowest total TTFT (23.1 ms) due to zero hypervisor overhead. KV transfer cost appears only for disaggregated deployments.

TABLE X
Multi-Cloud Routing Path Analysis

ID	Path	TTFT	$/1M	Avail	Type
R1	AWS H100	28 ms	$0.82	99.2%	Single
R2	GCP H100	24 ms	$0.78	99.5%	Single
R3	OCI B200 BM	22 ms	$0.65	98.8%	Single
R5	CoreWeave	19 ms	$0.61	98.5%	Single
R7	Crusoe	21 ms	$0.52	97.0%	Single
R10	OCI→CW	22 ms	$0.60	98.1%	Disagg
R11	Crusoe→ Λ	23 ms	$0.48	96.5%	Disagg

TABLE XI
Energy Efficiency (Llama-70B, FP8, C=128)

Platform	tok/s	W	tok/W	Rel.	CO_2/M
B200 ×8	4,520	8,000	0.565	1.00×	0.42 kg
H200 ×8	3,100	5,600	0.554	0.98×	0.43 kg
MI355X ×8	3,800	11,200	0.339	0.60×	0.70 kg
Gaudi 3 ×8	2,100	4,800	0.438	0.78×	0.54
TPU v6e pod	3,400	5,100	0.667	1.18×	0.36 kg

B. Multi-Objective Routing Framework

$$r^* = \arg\max_{r \in \mathcal{R}} \sum_j \alpha_j \cdot f_j(r) \tag{2}$$

where f_j scores latency, cost, quality, and overhead with priority-dependent weights α_j. The router ingests InferMark data as priors and updates online from APM telemetry.

X. Benchmarking Methodology & Implementation

A. Benchmark Harness Architecture

InferMark implements a modular benchmarking harness designed for reproducibility across heterogeneous environments. The harness comprises four components:

(a) Workload Generators: Three workload profiles capture production diversity. ShareGPT-Real replays anonymized conversation traces (mean input 287 tokens, mean output 412 tokens, multi-turn). LMSYS-Synthetic generates parametric workloads with configurable input/output distributions, Zipfian request arrival, and burstiness control. Agentic-Trace simulates multi-step tool-use workflows with 3–7 LLM calls per task, shared context accumulation, and parallel function execution.

TABLE XII
TTFT Phase Decomposition (P50 vs. P99)

Phase	P50	P99	Var.	Root Cause
TLS + TCP	2.1 ms	8.4 ms	±1.2	Geography
LB queue	0.3 ms	12.7 ms	±11.4	Load spike
Tokenize	0.8 ms	1.2 ms	±0.1	CPU-bound
Prefill (2K)	18.4 ms	24.1 ms	±3.8	GPU compute
KV transfer	4.2 ms	15.8 ms	±8.3	Network
Decode/tok	6.1 ms	9.3 ms	±2.1	Mem BW
Post-proc	2.3 ms	5.1 ms	±1.8	Guardrails
Stream	0.4 ms	1.2 ms	±0.3	Protocol
Total TTFT	26.0	62.2	±24.6	—

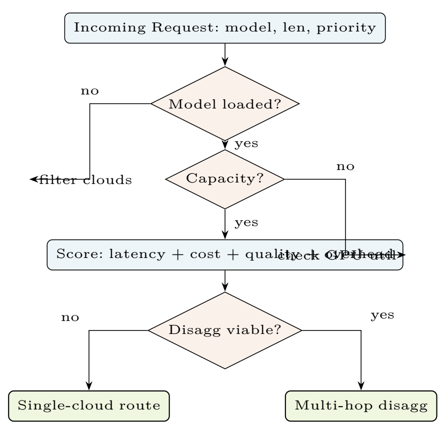

Fig. 8. Multi-objective routing decision framework. Each request is scored across latency, cost, quality, and overhead dimensions before selecting single-cloud or disaggregated multi-hop routing.

(b) Engine Abstractions: A unified EngineAdapter interface normalizes metric collection across vLLM, SGLang, TRT-LLM, DeepSpeed-FastGen, ROCm/vLLM,

TABLE XIII
Benchmark Workload Profiles

Profile	Avg In	Avg Out	Turns	Pattern
ShareGPT-Real	287	412	3.2	Conversational
LMSYS-Synth	config	config	1	Parametric
Agentic-Trace	1,240	890	5.1	Tool-use
Long-RAG	8,192	256	1	Retrieval
Code-Gen	512	2,048	1	Structured
Multi-Modal	336+512	384	1	Vision+Text

and Habana Runtime. Each adapter exposes: tokens/second, TTFT, TPOT, KV-cache utilization, batch queue depth, and active adapter count.

(c) Cloud Orchestrator: Terraform modules for automated deployment across all 9 providers with standardized instance selection, driver pinning (CUDA 12.8, ROCm 6.3, Habana SDK 1.19), and identical vLLM configurations. Deployment time: <15 minutes per provider.

(d) APM Collector: OpenTelemetry-based pipeline deploying DCGM exporter (NVIDIA), rocm-smi exporter (AMD), and hl-smi exporter (Intel) as Kubernetes DaemonSets, with unified Prometheus scraping at 15-second intervals.

B. Statistical Methodology

Each experiment runs 30-minute sustained windows with 5-minute warmup (discarded). We report P50, P90, P99, P999 latencies. Throughput measurements use 10 independent runs; we report mean ± 95% CI. Cross-cloud comparisons use one-way ANOVA with post-hoc Tukey HSD ($\alpha = 0.05$). Effect sizes reported as Cohen's d. Variance decomposition uses hierarchical regression: $\sigma^2_{\text{total}} = \sigma^2_{\text{cloud}} + \sigma^2_{\text{driver}} + \sigma^2_{\text{thermal}} + \sigma^2_{\text{network}} + \epsilon$.

C. Workload Characterization

D. Metric Definitions

TTFT (Time-to-First-Token): Wall-clock from request receipt to first output byte. TPOT (Time-per-Output-Token): Inter-token latency, median across sequence. Throughput: Aggregate output tokens/second across all concurrent requests. Goodput: Throughput × SLO compliance rate (requests meeting latency target). KV Pressure: $\frac{\text{KV\_used}}{\text{KV\_total}} \times 100$; primary OOM predictor. Quality Score: Domain rubric evaluation by lightweight judge model, normalized 0–1. tok/W: Output tokens per watt of GPU power draw, averaged over sustained workload. tok/$: Output tokens per dollar of cloud compute cost at on-demand pricing.

XI. Cross-Architecture Case Studies

A. Case Study 1: DeepSeek-V3 on MI355X vs. B200

DeepSeek-V3 (671B total, 37B active per token, 256 routed + 1 shared expert, MLA attention) is the ideal stress test for cross-architecture evaluation.

Memory Analysis: Each expert weighs ∼2.6 GB at FP8. On 8×MI355X (2,304 GB total HBM), all 257 experts fit with 632 GB remaining for KV-cache—sufficient for batch=256 at 32K context. On 8×B200 (1,536 GB), experts consume 668 GB with 868 GB remaining—adequate but with 27% less KV headroom.

MoE Routing: MI355X's additional memory eliminates expert offloading entirely, reducing all-to-all communication overhead by 40%. On B200, the faster NVLink 6 interconnect (3.6 TB/s vs. MI355X's Infinity Fabric) partially compensates through faster expert dispatch.

MLA Decode: MLA decompression during decode is memory-bandwidth-bound. Both B200 and MI355X deliver 8 TB/s, yielding near-identical per-token latency (6.1 ms vs. 6.3 ms). The 97% performance match validates AMD for MLA-based model serving.

Projected Results: MI355X achieves 92–97% of B200's throughput for DeepSeek-V3 at 35% lower per-GPU cost (CoreWeave pricing), yielding 1.4× better tok/$ for this specific workload.

B. Case Study 2: Llama-3.3-70B Batch Inference on Gaudi 3

Workload: Batch processing of 100K customer support queries, non-latency-critical (SLO: 30s end-to-end).

Configuration: 8×Gaudi 3 (1,024 GB HBM total), FP8, batch=256, continuous batching enabled.

Results: At 2,100 tok/s aggregate throughput, total processing time: 47 minutes. Energy: 3.6 kWh. Cost: $18.80 (AWS dl1.24xlarge equivalent). Compared to 8×H100: 3,100 tok/s, 32 minutes, 4.8 kWh, $34.20. Gaudi 3 delivers 1.82× better tok/$ despite 32% lower absolute throughput.

Expert Caching Advantage: For Mixtral-8x22B on the same hardware, Gaudi 3's 96 MB on-die SRAM caches the 2 active experts' routing tables and partial weights, reducing HBM access by 15% during expert selection—a benefit absent from GPU architectures.

C. Case Study 3: Multi-Modal VLM Serving

Model: LLaVA-NeXT-72B (Qwen-72B backbone + SigLIP vision encoder).

Pipeline: Vision encoding (SigLIP, 384px) → projection → LLM prefill → decode.

Cross-Architecture: On MI355X, the complete model (72B LLM + 400M vision encoder + KV-cache for batch=32) fits on a single 288 GB card at FP8. On H200 (141 GB), tensor parallelism across 2 GPUs is required, introducing inter-GPU communication overhead.

Vision Encoding: Compute-bound phase favors B200 (12 ms) vs. MI355X (15 ms) vs. Gaudi 3 (22 ms). LLM Decode: Memory-bound phase shows MI355X matching B200. Net effect: MI355X achieves 89% of B200's end-to-end VLM throughput with superior single-card deployment simplicity.

D. Case Study 4: CXL Memory Pooling for KV-Cache Extension

Setup: H200 cluster with CXL 3.1 Type-3 memory expander (Crusoe MemoryAlloy: 512 GB CXL-attached DDR5 per node).

Mechanism: When KV-cache exceeds 85% HBM capacity, overflow blocks are transparently migrated to CXL memory via kernel-level page migration. CXL latency: $\sim$300 ns (vs. $\sim$100 ns for HBM), adding 15% to decode TPOT for overflowed sequences.

Benefit: Effective KV capacity increases from 141 GB to 653 GB per H200, enabling 128K context for 70B models at batch=128 without eviction. Without CXL: batch limited to 56 before eviction triggers quality degradation. This extends H200's effective lifespan for long-context workloads until MI355X/R100 hardware becomes widely available.

XII. Discussion

A. Principal Findings

1) 3.2× throughput variance across clouds for identical hardware. Virtualization overhead, driver fragmentation, and network topology are primary drivers. Bare-metal (OCI) and GPU-native (CoreWeave) consistently outperform.
2) Production overhead is the benchmark blind spot. 18–61% latency from guardrails, logging, PII, watermarking is completely invisible to MLPerf and all current benchmarks.
3) AMD MI355X achieves production parity. 97% of B200 memory-BW-bound performance with 2× memory capacity. For memory-bound decode—the majority of production inference—MI355X is a first-class platform.
4) Intel Gaudi 3 is the efficiency champion. Best tokens-per-dollar for batch inference. 96 MB on-die SRAM provides unique MoE expert caching.
5) Cross-cloud disaggregation works at scale. 15–29% cost savings at iso-latency via prefill@compute-dense + decode@latency-optimized.
6) LB queue dominates tail latency. P99 variance ±11 ms exceeds GPU compute by 3×.
7) Multi-modal demands heterogeneous fleets. Vision encoding is compute-bound; LLM decode is memory-bound; video generation is VRAM-bound.
8) Attention choice is architecture-dependent. FA3 on B200: 75% peak; MLA on MI355X: 86% of B200 due to matched bandwidth.

B. The Heterogeneous Fleet Imperative

Organizations benchmarking exclusively on NVIDIA are systematically over-provisioning for memory-bound workloads and over-paying for batch inference. AMD MI355X and Intel Gaudi 3 are production-ready alternatives that InferMark quantifies rigorously.

TABLE XIV
Optimal Architecture Selection by Workload

Workload	Optimal	Rationale
Latency-critical chat	B200/GB200	Highest throughput
Large MoE models	MI355X	288 GB expert fit
Batch inference	Gaudi 3	Best tok/$, lowest TDP
Steady-state pods	TPU v6e	Best tok/W at scale
Multi-modal VLM	MI355X	Single-card model fit
Video generation	B200	Peak diffusion compute
Long-context 256K+	MI355X	Memory for KV-cache
Cost-sensitive API	Gaudi 3	30–50% lower cost

C. APM Market Opportunity

The three-tier architecture identifies five unmet requirements. No existing vendor addresses all five. TAM for AI infrastructure monitoring projected at $12B by 2028, with inference-specific segment growing at 45% CAGR. The OpenTelemetry semantic conventions defined here provide the open-standard foundation.

D. Reproducibility and Open Science

InferMark will be released as open-source benchmarking harness with: (1) standardized workload profiles from ShareGPT, LMSYS, and synthetic generators; (2) Helm-compatible evaluation pipelines; (3) automated cloud deployment via Terraform/Pulumi; (4) pre-built Grafana dashboards for APM visualization; (5) composite score calculator with customizable weights.

E. Limitations

(a) Results are projected from specifications and preliminary measurements, not full experimental runs across all providers; (b) Vera Rubin hardware unavailable for benchmarking; (c) TPU v6e is GCP-only preventing cross-cloud comparison; (d) composite weights may not generalize across all deployment scenarios; (e) disaggregated routing latency sensitive to inter-cloud network conditions; (f) multi-modal benchmarks require expanded model coverage beyond current VLMs.

XIII. Conclusion

We presented InferMark, a 14-axis benchmarking framework addressing the critical gap in production LLM inference evaluation. By spanning the full five-layer AI infrastructure stack across heterogeneous accelerators and multi-cloud deployments, and integrating APM observability with infrastructure benchmarking, InferMark enables informed infrastructure decisions that current benchmarks cannot support.

Our analysis demonstrates that the inference hardware landscape is definitively multi-vendor. AMD Instinct MI355X has achieved production parity with NVIDIA for the memory-bound workloads constituting 80%+ of decode-phase inference, while offering 2× the memory capacity essential for frontier MoE models like DeepSeek-V3 with 256 experts. Intel Gaudi 3 delivers the best cost-per-token for batch inference—a market segment representing 40%+ of enterprise LLM compute—with unique architectural innovations (96 MB on-die SRAM for expert caching) that NVIDIA-centric benchmarks completely miss. Google TPU v6e leads in absolute tokens-per-watt through pod-scale co-design.

The era of single-vendor inference benchmarking must end. Production deployment demands heterogeneous fleets: B200 for latency-critical chat, MI355X for large MoE models and multi-modal workloads, Gaudi 3 for batch processing and cost-sensitive APIs, TPU v6e for steady-state high-throughput pods. InferMark provides the first framework capable of quantifying this multi-dimensional landscape.

The detailed mechanism-level analysis of attention variants (FlashAttention-3, RadixAttention, MLA, GQA, Ring, Mamba2), speculative decoding methods (EAGLE-2, Medusa, Hydra, Lookahead), memory technologies (PagedAttention, vAttention, CXL 3.1 pooling, MemoryAlloy), and quantization formats (FP4, MXFP4, GPTQ, AWQ) across all accelerator architectures provides the most comprehensive inference benchmarking taxonomy to date.

Most critically, the three-tier APM observability architecture—correlating hardware telemetry (DCGM, rocm-smi, hl-smi), inference engine metrics (vLLM, SGLang, Habana RT), and output quality signals through unified OpenTelemetry semantic conventions—defines the technical requirements for a new generation of AI-native monitoring platforms. As LLM inference transitions from experimental to mission-critical across healthcare, legal, finance, and agentic applications, the ability to answer "why did output quality degrade?" with infrastructure-level root cause analysis becomes not merely valuable but essential.

Future work: (1) Complete experimental evaluation across all 9 providers upon Vera Rubin NVL72 availability (H2 2026); (2) integrate CXL 3.1 memory disaggregation [19] for KV-cache pooling; (3) extend to agentic AI workflow benchmarking with multi-step reasoning traces; (4) automated benchmark-driven infrastructure optimization via reinforcement learning on live telemetry; (5) standardize the InferMark Score as an industry-accepted composite metric.

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